# BEAM DESIGN CALCULATION

| | | |
|---|---|---|
| Project Name | : | G+2 RB BEAM DESIGN @MAIN ROOF |
| Client Name | : | Unassigned |
| Engineer Name | : | Unassigned |
| Design File | : | E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-11.125 m.R1.rcdx |
| Analysis File | : | E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std |
| Analysis Last Modified | : | 06-10-2026 00:54:36 |
| Level Designed | : | 11.125 m |

| Sr.No. | Symbol | | Definitions |
|---|---|---|---|
| 1 | α | = | Angle formed with horizontal by diagonal reinforcement |
| 2 | ε1 | = | Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone |
| 3 | εm | = | Average steel strain at level considered |
| 4 | Acr | = | Distance from the point considered to the surface of the nearest longitudinal bar |
| 5 | Adr | = | Area of diagonal reinforcement in coupler beam |
| 6 | AstCalc | = | Area of Tension reinforcement required in sqmm |
| 7 | Ast comp | = | Ast at compression side |
| 8 | AstCrack | = | Area of Tension reinforcement for Crack Width required in sqmm |
| 9 | AstPrv | = | Area of reinforcement provided at given section in sqmm |
| 10 | Asv | = | Area of shear reinforcement required per meter length in sqmm |
| 11 | Asv Reqd | = | Min area shear reinforcement as per clause 41.4.3 |
| 12 | Asv Total Prv | = | Total area of shear reinforcement provided, including that for torsion |
| 13 | Asv Torsion | = | Area of Stirrups over the corner bars to resist shear and torsion as per clause 41.4.3 |
| 14 | Asv Torsion Prv | = | Area of shear reinforcement resisting torsion (outer legs only) |
| 15 | b | = | Width of the Beam in mm |
| 16 | bw | = | Width of Web in mm |
| 17 | Ceff | = | Effective Cover of Beam in mm |
| 18 | Cmin | = | Clear cover to top and bottom reinforcement in mm |
| 19 | Cside | = | Clear cover to side reinforcement in mm |
| 20 | d | = | Effective depth of Beam in mm |
| 21 | d' | = | Clear cover to reinforcement at compression side. |
| 22 | D | = | Depth of Beam in mm |
| 23 | D' | = | Distance (c/c) between top-most and bottom-most layer |
| 24 | Df | = | Thickness of Flange in mm |
| 25 | Ec | = | Modulus of elasticity of concrete |
| 26 | Es | = | Modulus of elasticity of steel |
| 27 | FcPerm | = | Permissible Stress in Concrete required in N/sqmm |

| | | | |
|---|---|---|---|
| 28 | Fst | = | Stress in steel |
| 29 | FstPerm | = | Permissible Stress in Steel required in N/sqmm |
| 30 | Legs | = | Number of legs of the shear reinforcement |
| 31 | lo | = | Effective Length of Beam in mm |
| 32 | Mh | = | Hogging Moment of Resistance in kNm |
| 33 | Ms | = | Sagging Moment of Resistance in kNm |
| 34 | $M_{Tu}$ | = | Converted Bending Moment due to Torsion at a section in kNm |
| 35 | Mu | = | Factored Bending Moment in Limit State at a section in kNm |
| 36 | Mud | = | Total Design Moment = Mu + $M_{Tu}$ in kNm |
| 37 | MuLim | = | Limiting Moment of Resistence of Singly Reinforced Section |
| 38 | NA angle (φ) | = | It is a neutral axis angle corresponding to load angle to find out MCap |
| 39 | Pcclc | = | Calculated percentage compression steel |
| 40 | Ptclc | = | Calculated percentage tension steel |
| 41 | Ptmin | = | Minimum percentage steel as per code |
| 42 | PtPrv | = | provided percentage steel |
| 43 | R | = | Moment Ratio Mud/$Bd^2$ |
| 44 | Reinf | = | Rebar arrangement provided |
| 45 | sp | = | Spacing Between bars at outer most layer |
| 46 | Spc1 to Spc6 | = | Max Stirrup Criteria in mm |
| 47 | Stirrup Dia | = | Diameter of shear stirrups in mm |
| 48 | SvCalc | = | Stirrup spacing calculated as per Asv in mm |
| 49 | SvPrv | = | Stirrup spacing provided in m |
| 50 | Tc | = | Permissible Shear stress in concrete at a given section in N/sqmm Calculated as per Tensile reinforcement |
| 51 | Tce | = | Earthquake inducted shear stress in N/sqmm |
| 52 | Tcep | = | Permissible value of shear stress for checking of coupling action in N/sqmm |
| 53 | Tu | = | Factored Torsional Moment in Limit State at a section in kNm |
| 54 | Tv | = | Shear stress due to Vud in N/sqmm |
| 55 | $V^{D+L}$ | = | Shear force 1.2x(Dead Load + Live Load) in kN |
| 56 | Vc | = | Shear capacity of concrete section due to Tc in kN |
| 57 | Vu | = | Factored Shear Force in Limit State at a section in kN |
| 58 | Vu-A1 | = | $V^{D+L}$Left - 1.4 x (MsLeft + MhRight)/L in kN |
| 59 | Vu-A2 | = | $V^{D+L}$Left + 1.4 x (MhLeft + MsRight)/L in kN |
| 60 | Vu-B1 | = | $V^{D+L}$Right + 1.4 x (MsLeft + MhRight)/L in kN |
| 61 | Vu-B2 | = | $V^{D+L}$Right - 1.4 x (MhLeft + MsRight)/L in kN |
| 62 | Vud | = | Design Shear Force (Max of Vus and Vu-sway) in kN |
| 63 | Vue | = | Earthquake inducted shear in kN |
| 64 | Vu-sway | = | Max of (Vu-A1 and Vu-A2) & (Vu-B1 and Vu-B2) in kN |
| 65 | Vus | = | Vud - Vc Design Shear Force for reinforcement, in kN |
| 66 | Vut | = | Total Shear Force = Vu + $V_{Tu}$ in kN |

| | | | |
|---|---|---|---|
| 67 | $V_{Tu}$ | = | Converted Shear Force due to Torsion at a section in kN |
| 68 | Wcr | = | Surface Crack Width |
| 69 | WcrPerm | = | Permissible Crack Width required in mm |
| 70 | X1 | = | Smaller c/c dimension of outermost Stirrups |
| 71 | Y1 | = | Larger c/c dimension of outermost Stirrups |
| | | | All 6mm diameter bars are assumed 250 grade with fy = 250 N/sqmm |

All Forces are in 'kN', 'kNm', Stress in 'N/sqmm' & Dimension are in 'mm'.

**Code References**

**IS 456**

| Sr.No. | Item | | Clause / Table |
|---|---|---|---|
| 1. | Ptmax | : | 26.5.1.1 |
| 2. | Ptmin | : | 26.5.1.1 |
| 3. | $M_{Tu}$ | : | 41.4.2 |
| 4. | $V_{Tu}$ | : | 41.3.1 |
| 5. | Ptclc | : | 38 |
| 6. | Tc | : | 40.2.1 |
| 7. | Tcmax | : | 40.2.3 |
| 8. | Asv | : | 40.4 |
| 9. | Min Shear Reinf | : | 26.5.1.6 |
| 10. | Max Stirrup Spacing | : | 26.5.1.5 |
| 11. | Shear Reinf - Torsion | : | 26.5.1.7 |
| 12. | Side Face Reinforcement | : | 26.5.1.3 |
| 13. | crack width calculation | : | Annex-F |
| 14. | cracking | : | Clause 35.3.2 |

**IS 13920 - 2016**

| Sr.No. | Item | | Clause / Table |
|---|---|---|---|
| 1. | Ptmax | : | 6.2.2 |
| 2. | Ptmin | : | 6.2.1 |
| 3. | Astmin | : | 6.2.3 & 6.2.4 |
| 4. | Svclc | : | 6.3.5 |
| 5. | Shear Capacity | : | 6.3.4 |
| 6. | Coupling Beams | : | 10.5 |

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| | | | |
|---|---|---|---|
| Group | : | **G1** | |
| Beam No | : | **B1** | |
| Analysis Reference (Member) | : | 7 (11.125 m) | |
| Beam Length | : | 3924 | mm |

| | | | |
|---|---|---|---|
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 14 | 14 | 19 | 12 |
| Critical L/C - RCDC | 8 | 8 | 10 | 10 | 15 | 8 |
| Mu (kNm) | 26.3 | 24.76 | 10.95 | 24.44 | 9.29 | 21.87 |
| Tu (kNm) | 0.66 | 0.66 | 0.39 | 0.39 | 0.18 | 0.66 |
| $M_{Tu}$ (kNm) | 1.12 | 1.12 | 0.65 | 0.65 | 0.3 | 1.12 |
| Mud (kNm) | 27.42 | 25.88 | 11.6 | 25.09 | 9.59 | 22.99 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.871 | 0.822 | 0.368 | 0.797 | 0.305 | 0.73 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |

D = AstCrack = Area of reinforcement required for Crackwidth

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 14 | 12 | 12 |
| Critical L/C - RCDC | 10 | 8 | 8 |
| PtPrv (%) | 1.107 | 0.266 | 1.107 |
| Vu (kN) | 23.07 | 17.08 | 26.89 |
| Tu (kNm) | 0.39 | 0.66 | 0.66 |
| $V_{Tu}$ (kN) | 2.69 | 4.6 | 4.6 |
| Vut (kN) | 25.76 | 21.68 | 31.49 |
| Tv (N/sqmm) | 0.3 | 0.25 | 0.37 |
| Tc (N/sqmm) | 0.67 | 0.37 | 0.67 |
| Vc (kN) | 56.69 | 31.88 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 93.435 | 88.94 | 120.007 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

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| | | |
|---|---|---|
| Group | : | **G1** |
| Beam No | : | **B2** |
| Analysis Reference (Member) | : | 8 (11.125 m) |

| | | | |
|---|---|---|---|
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 17 | 19 | 19 | 12 | 12 |
| Critical L/C - RCDC | 8 | 13 | 15 | 15 | 8 | 8 |
| Mu (kNm) | 29.95 | 6.02 | 2.42 | 18.34 | 64.58 | 110.52 |
| Tu (kNm) | 0.4 | 0.19 | 0.42 | 0.42 | 0.4 | 0.4 |
| $M_{Tu}$ (kNm) | 0.67 | 0.32 | 0.71 | 0.71 | 0.67 | 0.67 |
| Mud (kNm) | 30.62 | 6.34 | 3.13 | 19.05 | 65.24 | 111.19 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.972 | 0.201 | 0.099 | 0.605 | 2.072 | 3.531 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.213 | 0.2 | 0.2 | 0.2 | 0.485 | 0.903 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0.082 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 181.49 | 170.2 | 170.2 | 170.2 | 412.78 | 768.43 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |

| | | | | |
|---|---|---|---|---|
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 12 |
| Critical L/C - RCDC | 8 | 8 | 8 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 30.9 | 40.71 | 50.53 |
| Tu (kNm) | 0.4 | 0.4 | 0.4 |
| $V_{Tu}$ (kN) | 2.76 | 2.76 | 2.76 |
| Vut (kN) | 33.65 | 43.47 | 53.28 |
| Tv (N/sqmm) | 0.4 | 0.51 | 0.63 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 118.696 | 149.763 | 180.83 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

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Group : **G1**

Beam No : **B3**

| | | | |
|---|---|---|---|
| Analysis Reference (Member) | : | 142 (11.125 m) | |
| Beam Length | : | 1372 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 15 | 15 | 15 | 12 | 12 | 15 |
| Critical L/C - RCDC | 11 | 11 | 11 | 8 | 8 | 11 |
| Mu (kNm) | - | - | 5.08 | 91.3 | 65.89 | 14.08 |
| Tu (kNm) | 9.74 | 9.74 | 9.74 | 6.01 | 6.01 | 9.74 |
| $M_{Tu}$ (kNm) | 16.44 | 16.44 | 16.44 | 10.15 | 10.15 | 16.44 |
| Mud (kNm) | 16.44 | 16.44 | 21.52 | 101.44 | 76.04 | 30.52 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.522 | 0.522 | 0.684 | 3.222 | 2.415 | 0.969 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.823 | 0.579 | 0.213 |
| Pcclc (%) (C) | 0.112 | 0.112 | 0.077 | 0 | 0 | 0.016 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 700.45 | 492.5 | 180.9 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural

| | | | | |
|---|---|---|---|---|
| | | | | reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 12 |
| Critical L/C - RCDC | 8 | 8 | 8 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 75.67 | 70.86 | 63.12 |
| Tu (kNm) | 6.01 | 6.01 | 6.01 |
| $V_{Tu}$ (kN) | 41.82 | 41.82 | 41.82 |
| Vut (kN) | 117.49 | 112.68 | 104.94 |
| Tv (N/sqmm) | 1.38 | 1.32 | 1.23 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 60.8 | 55.99 | 48.25 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 556.671 | 541.444 | 516.961 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 455.16 | 419.15 | 361.25 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

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| | | | |
|---|---|---|---|
| Group | : | **G2** | |
| Beam No | : | **B4** | |
| Analysis Reference (Member) | : | 27 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 14 | 19 | 19 | 7 |
| Critical L/C - RCDC | 8 | 8 | 10 | 15 | 15 | 3 |
| Mu (kNm) | 33.03 | 21.06 | 18.4 | 22.64 | 12.56 | 17.92 |
| Tu (kNm) | 0.15 | 0.15 | 0 | 0 | 0 | 0.1 |
| $M_{Tu}$ (kNm) | 0.25 | 0.25 | 0 | 0 | 0 | 0.17 |
| Mud (kNm) | 33.28 | 21.31 | 18.4 | 22.65 | 12.56 | 18.09 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 1.057 | 0.677 | 0.584 | 0.719 | 0.399 | 0.575 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.233 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 198.2 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

| | | | | |
|---|---|---|---|---|
| AstCalc | = | Max {B',C} x B X d (for Mu > 0 ) | | |
| AstCalc | = | Bn x B X d (for Mu = 0) | | |
| Where, | | | | |
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 9 | 7 | 7 |
| Critical L/C - RCDC | 5 | 3 | 3 |
| PtPrv (%) | 1.107 | 0.266 | 1.107 |
| Vu (kN) | 13.74 | 15.02 | 17.73 |
| Tu (kNm) | 0 | 0.1 | 0.1 |
| $V_{Tu}$ (kN) | 0 | 0.71 | 0.71 |
| Vut (kN) | 13.74 | 15.73 | 18.44 |
| Tv (N/sqmm) | 0.16 | 0.18 | 0.22 |
| Tc (N/sqmm) | 0.67 | 0.37 | 0.67 |
| Vc (kN) | 56.69 | 31.88 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | - | 52.945 | 61.517 |
| Asv Torsion Prv (sqmm) | - | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |

| | | |
|---|---|---|
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |
| Side Face Reinforcement not required. | | |

---

| | | | |
|---|---|---|---|
| Group | : | **G2** | |
| Beam No | : | **B5** | |
| Analysis Reference (Member) | : | 28 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 17 | 19 | 19 | 12 | 12 |
| Critical L/C - RCDC | 8 | 13 | 15 | 15 | 8 | 8 |
| Mu (kNm) | 39.38 | 4.73 | 1.88 | 13.53 | 69.52 | 105.82 |
| Tu (kNm) | 0.54 | 0.37 | 0.16 | 0.16 | 0.54 | 0.54 |
| $M_{Tu}$ (kNm) | 0.91 | 0.62 | 0.27 | 0.27 | 0.91 | 0.91 |
| Mud (kNm) | 40.3 | 5.35 | 2.15 | 13.8 | 70.44 | 106.74 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 1.28 | 0.17 | 0.068 | 0.438 | 2.237 | 3.39 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.285 | 0.2 | 0.2 | 0.2 | 0.529 | 0.868 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0.04 |
| PtPrv (%) | 0.399 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 242.95 | 170.2 | 170.2 | 170.2 | 450.59 | 738.41 |
| AstPrv (sqmm) | 339.3 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 3-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

| | | | | |
|---|---|---|---|---|
| AstCalc | = | Max {B',C} x B X d (for Mu > 0 ) | | |
| AstCalc | = | Bn x B X d (for Mu = 0) | | |
| Where, | | | | |
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 7 |
| Critical L/C - RCDC | 8 | 8 | 3 |
| PtPrv (%) | 0.266 | 1.107 | 1.107 |
| Vu (kN) | 37.01 | 37.01 | 37.88 |
| Tu (kNm) | 0.54 | 0.54 | 0.4 |
| $V_{Tu}$ (kN) | 3.77 | 3.77 | 2.76 |
| Vut (kN) | 40.78 | 40.78 | 40.64 |
| Tv (N/sqmm) | 0.48 | 0.48 | 0.48 |
| Tc (N/sqmm) | 0.37 | 0.67 | 0.67 |
| Vc (kN) | 31.88 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 8.9 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 145.713 | 145.713 | 140.832 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G2** | |
| Beam No | : | **B6** | |
| Analysis Reference (Member) | : | 147 (11.125 m) | |
| Beam Length | : | 1372 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 13 | 13 | 13 | 5 | 12 | 13 |
| Critical L/C - RCDC | 9 | 9 | 9 | 1 | 8 | 9 |
| Mu (kNm) | - | - | 12.17 | 98.76 | 70.86 | 12 |
| Tu (kNm) | 8.27 | 8.27 | 8.27 | 4.61 | 4.53 | 8.27 |
| $M_{Tu}$ (kNm) | 13.97 | 13.97 | 13.97 | 7.77 | 7.65 | 13.97 |
| Mud (kNm) | 13.97 | 13.97 | 26.14 | 106.53 | 78.5 | 25.97 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.444 | 0.444 | 0.83 | 3.383 | 2.493 | 0.825 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.866 | 0.601 | 0.2 |
| Pcclc (%) (C) | 0.095 | 0.095 | 0.012 | 0.038 | 0 | 0.013 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 737.01 | 511.33 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

| | | | | |
|---|---|---|---|---|
| AstCalc | = | Max {B',C} x B X d (for Mu > 0 ) | | |
| AstCalc | = | Bn x B X d (for Mu = 0) | | |
| Where, | | | | |
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 1.107 | 1.107 | 0.266 |
| Vu (kN) | 85.6 | 83.8 | 76.49 |
| Tu (kNm) | 4.61 | 4.61 | 4.61 |
| $V_{Tu}$ (kN) | 32.04 | 32.04 | 32.04 |
| Vut (kN) | 117.64 | 115.84 | 108.53 |
| Tv (N/sqmm) | 1.38 | 1.36 | 1.28 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.37 |
| Vc (kN) | 56.69 | 56.69 | 31.88 |
| Vus=Vut-Vc (kN) | 60.95 | 59.15 | 76.65 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 513.926 | 508.25 | 485.094 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 456.27 | 442.84 | 573.79 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G3** | |
| Beam No | : | **B7** | |
| Analysis Reference (Member) | : | 47 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 19 | 19 | 19 | 7 |
| Critical L/C - RCDC | 8 | 8 | 15 | 15 | 15 | 3 |
| Mu (kNm) | 30.92 | 17.81 | 18.64 | 22.8 | 12.44 | 24.09 |
| Tu (kNm) | 0.32 | 0.32 | 0.34 | 0.34 | 0.34 | 0.33 |
| $M_{Tu}$ (kNm) | 0.54 | 0.54 | 0.57 | 0.57 | 0.57 | 0.55 |
| Mud (kNm) | 31.45 | 18.34 | 19.21 | 23.37 | 13.01 | 24.64 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.999 | 0.583 | 0.61 | 0.742 | 0.413 | 0.783 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ptclc (%) (B') | 0.219 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 186.72 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 9 | 7 | 7 |
| Critical L/C - RCDC | 5 | 3 | 3 |
| PtPrv (%) | 0.709 | 0.709 | 0.709 |
| Vu (kN) | 13.5 | 16.24 | 18.95 |
| Tu (kNm) | 0.47 | 0.32 | 0.32 |
| $V_{Tu}$ (kN) | 3.28 | 2.26 | 2.26 |
| Vut (kN) | 16.77 | 18.51 | 21.22 |
| Tv (N/sqmm) | 0.2 | 0.22 | 0.25 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.56 |
| Vc (kN) | 47.85 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 66.171 | 67.371 | 75.846 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G3** | |
| Beam No | : | **B8** | |
| Analysis Reference (Member) | : | 48 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 17 | 19 | 19 | 12 | 12 |
| Critical L/C - RCDC | 8 | 13 | 15 | 15 | 8 | 8 |
| Mu (kNm) | 29.32 | 7.22 | 7.48 | 15.57 | 39.54 | 62.49 |
| Tu (kNm) | 0.54 | 0.37 | 0.15 | 0.15 | 0.54 | 0.54 |
| $M_{Tu}$ (kNm) | 0.91 | 0.62 | 0.25 | 0.25 | 0.91 | 0.91 |
| Mud (kNm) | 30.23 | 7.85 | 7.73 | 15.82 | 40.45 | 63.4 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.96 | 0.249 | 0.246 | 0.503 | 1.285 | 2.014 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.21 | 0.2 | 0.2 | 0.2 | 0.287 | 0.47 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 179.05 | 170.2 | 170.2 | 170.2 | 243.92 | 399.58 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 7 |
| Critical L/C - RCDC | 8 | 8 | 3 |
| PtPrv (%) | 0.266 | 0.709 | 0.709 |
| Vu (kN) | 23.4 | 23.4 | 26.39 |
| Tu (kNm) | 0.54 | 0.54 | 0.5 |
| $V_{Tu}$ (kN) | 3.75 | 3.75 | 3.5 |
| Vut (kN) | 27.15 | 27.15 | 29.89 |
| Tv (N/sqmm) | 0.32 | 0.32 | 0.35 |
| Tc (N/sqmm) | 0.37 | 0.56 | 0.56 |
| Vc (kN) | 31.88 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 100.64 | 100.64 | 108.146 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G3** | |
| Beam No | : | **B9** | |
| Analysis Reference (Member) | : | 152 (11.125 m) | |
| Beam Length | : | 1372 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 13 | 13 | 13 | 5 | 13 | 13 |
| Critical L/C - RCDC | 9 | 9 | 9 | 1 | 9 | 9 |
| Mu (kNm) | - | - | - | 56.26 | 36.06 | 14.02 |
| Tu (kNm) | 8.81 | 8.81 | 8.81 | 4.84 | 8.81 | 8.81 |
| $M_{Tu}$ (kNm) | 14.88 | 14.88 | 14.88 | 8.17 | 14.88 | 14.88 |
| Mud (kNm) | 14.88 | 14.88 | 14.88 | 64.43 | 50.94 | 28.9 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |

| | | | | | | |
|---|---|---|---|---|---|---|
| R | 0.472 | 0.472 | 0.472 | 2.046 | 1.618 | 0.918 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.478 | 0.368 | 0.201 |
| Pcclc (%) (C) | 0.101 | 0.101 | 0.101 | 0 | 0 | 0.006 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 406.94 | 313.18 | 170.8 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.709 | 0.709 | 0.709 |
| Vu (kN) | 39.89 | 39.48 | 38.19 |
| Tu (kNm) | 4.84 | 4.84 | 4.84 |
| $V_{Tu}$ (kN) | 33.66 | 33.66 | 33.66 |
| Vut (kN) | 73.55 | 73.14 | 71.85 |
| Tv (N/sqmm) | 0.86 | 0.86 | 0.84 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.56 |
| Vc (kN) | 47.85 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 25.7 | 25.29 | 24 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 370.674 | 369.379 | 365.338 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G4** | |
| Beam No | : | **B10** | |
| Analysis Reference (Member) | : | 67 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 19 | 14 | 19 | 7 |
| | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Critical L/C - RCDC | 8 | 8 | 15 | 10 | 15 | 3 |
| Mu (kNm) | 31.47 | 19.83 | 23.28 | 31.04 | 15.12 | 33.77 |
| Tu (kNm) | 0.22 | 0.22 | 0.15 | 0.24 | 0.15 | 0.29 |
| $M_{Tu}$ (kNm) | 0.38 | 0.38 | 0.25 | 0.4 | 0.25 | 0.49 |
| Mud (kNm) | 31.85 | 20.2 | 23.53 | 31.45 | 15.36 | 34.25 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 1.012 | 0.642 | 0.747 | 0.999 | 0.488 | 1.088 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.222 | 0.2 | 0.2 | 0.219 | 0.2 | 0.24 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 189.2 | 170.2 | 170.2 | 186.68 | 170.2 | 204.33 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 9 | 7 | 7 |
| Critical L/C - RCDC | 5 | 3 | 3 |
| PtPrv (%) | 0.709 | 0.709 | 0.709 |
| Vu (kN) | 21.77 | 20.6 | 26.91 |
| Tu (kNm) | 0.3 | 0.29 | 0.29 |
| $V_{Tu}$ (kN) | 2.1 | 2.02 | 2.02 |
| Vut (kN) | 23.88 | 22.61 | 28.93 |
| Tv (N/sqmm) | 0.28 | 0.27 | 0.34 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.56 |
| Vc (kN) | 47.85 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |

| | | | |
|---|---|---|---|
| Asv Torsion (sqmm) | 83.514 | 79.177 | 98.94 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G4** | |
| Beam No | : | **B11** | |
| Analysis Reference (Member) | : | 68 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

| **Flexure Design** | | | | | | |
|---|---|---|---|---|---|---|
| | **Beam Bottom** | | | **Beam Top** | | |
| | | | | | | |

| | Left | Mid | Right | Left | Mid | Right |
|---|---|---|---|---|---|---|
| Critical L/C - Analysis | 12 | 12 | 19 | 19 | 12 | 12 |
| Critical L/C - RCDC | 8 | 8 | 15 | 15 | 8 | 8 |
| Mu (kNm) | 33.25 | 13.14 | 6.66 | 20.18 | 35.12 | 63.26 |
| Tu (kNm) | 0.84 | 0.84 | 0.57 | 0.57 | 0.84 | 0.84 |
| $M_{Tu}$ (kNm) | 1.41 | 1.41 | 0.97 | 0.97 | 1.41 | 1.41 |
| Mud (kNm) | 34.66 | 14.55 | 7.63 | 21.14 | 36.53 | 64.67 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 1.101 | 0.462 | 0.242 | 0.672 | 1.16 | 2.054 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.243 | 0.2 | 0.2 | 0.2 | 0.257 | 0.48 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 206.89 | 170.2 | 170.2 | 170.2 | 218.76 | 408.7 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 12 | 7 | 7 |
| Critical L/C - RCDC | 8 | 3 | 3 |
| PtPrv (%) | 0.266 | 0.709 | 0.709 |
| Vu (kN) | 22.5 | 27.17 | 33.48 |
| Tu (kNm) | 0.84 | 0.93 | 0.93 |
| $V_{Tu}$ (kN) | 5.82 | 6.44 | 6.44 |
| Vut (kN) | 28.32 | 33.61 | 39.93 |
| Tv (N/sqmm) | 0.33 | 0.39 | 0.47 |
| Tc (N/sqmm) | 0.37 | 0.56 | 0.56 |
| Vc (kN) | 31.88 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |

| | | | |
|---|---|---|---|
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 112.93 | 132.077 | 151.841 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G4** | |
| Beam No | : | **B12** | |
| Analysis Reference (Member) | : | 157 (11.125 m) | |
| Beam Length | : | 1372 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 13 | 13 | 13 | 5 | 5 | 13 |
| Critical L/C - RCDC | 9 | 9 | 9 | 1 | 1 | 9 |
| Mu (kNm) | - | - | 8.53 | 64.05 | 45.06 | 7.12 |
| Tu (kNm) | 5.06 | 5.06 | 5.06 | 1.8 | 1.8 | 5.06 |
| $M_{Tu}$ (kNm) | 8.54 | 8.54 | 8.54 | 3.03 | 3.03 | 8.54 |
| Mud (kNm) | 8.54 | 8.54 | 17.07 | 67.08 | 48.1 | 15.66 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.271 | 0.271 | 0.542 | 2.131 | 1.527 | 0.497 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.501 | 0.346 | 0.2 |
| Pcclc (%) (C) | 0.057 | 0.057 | 0 | 0 | 0 | 0.009 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 426.07 | 294.14 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | **Left** | **Mid** | **Right** |
|---|---|---|---|
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.709 | 0.709 | 0.709 |
| Vu (kN) | 55.56 | 54.67 | 51.87 |
| Tu (kNm) | 1.8 | 1.8 | 1.8 |
| $V_{Tu}$ (kN) | 12.51 | 12.51 | 12.51 |
| Vut (kN) | 68.07 | 67.18 | 64.38 |
| Tv (N/sqmm) | 0.8 | 0.79 | 0.76 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.56 |

| | | | |
|---|---|---|---|
| Vc (kN) | 47.85 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 20.22 | 19.33 | 16.53 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 265.24 | 262.434 | 253.68 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G5** | |
| Beam No | : | **B13** | |
| Analysis Reference (Member) | : | 132 (11.125 m) | |
| Beam Length | : | 3924 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |

| | | | |
|---|---|---|---|
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 7 | 5 | 5 | 9 | 5 | 5 |
| Critical L/C - RCDC | 3 | 1 | 1 | 5 | 1 | 1 |
| Mu (kNm) | 10.61 | 15.5 | 6.29 | 11.95 | - | 16.34 |
| Tu (kNm) | 1.24 | 1.29 | 1.29 | 0.82 | 1.29 | 1.29 |
| $M_{Tu}$ (kNm) | 2.09 | 2.18 | 2.18 | 1.39 | 2.18 | 2.18 |
| Mud (kNm) | 12.7 | 17.68 | 8.47 | 13.34 | 2.18 | 18.52 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.403 | 0.561 | 0.269 | 0.424 | 0.069 | 0.588 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0.015 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | **Left** | **Mid** | **Right** |
|---|---|---|---|
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 |

| | | | |
|---|---|---|---|
| Vu (kN) | 24.37 | 8.35 | 27.95 |
| Tu (kNm) | 1.29 | 1.29 | 1.29 |
| $V_{Tu}$ (kN) | 8.98 | 8.98 | 8.98 |
| Vut (kN) | 33.35 | 17.33 | 36.93 |
| Tv (N/sqmm) | 0.39 | 0.2 | 0.43 |
| Tc (N/sqmm) | 0.37 | 0.37 | 0.37 |
| Vc (kN) | 31.88 | 31.88 | 31.88 |
| Vus=Vut-Vc (kN) | 1.47 | 0 | 5.05 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 138.602 | 89.039 | 149.677 |
| Asv Torsion Prv (sqmm) | 718.14 | 365.6 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 275 | 140 |
| SvPrv (mm) | 140 | 275 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 365.6 | 718.14 |

**Maximum Spacing Criteria**

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

**SFR Design**

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

---

Group : **G5**

Beam No : **B14**

Analysis Reference (Member) : 133 (11.125 m)

Beam Length : 3924 mm

Breadth (B) : 230 mm

Depth (D) : 430 mm

| | | | |
|---|---|---|---|
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 8 | 8 | 9 | 14 | 12 | 12 |
| Critical L/C - RCDC | 4 | 4 | 5 | 10 | 8 | 8 |
| Mu (kNm) | 6.26 | 9.84 | 2.96 | 12.47 | 5.25 | 24.78 |
| Tu (kNm) | 1.21 | 1.21 | 0.61 | 0.61 | 0 | 0 |
| $M_{Tu}$ (kNm) | 2.05 | 2.05 | 1.04 | 1.03 | 0 | 0 |
| Mud (kNm) | 8.31 | 11.88 | 4 | 13.5 | 5.25 | 24.78 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.264 | 0.377 | 0.127 | 0.429 | 0.167 | 0.787 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 12 | 5 |
| Critical L/C - RCDC | 1 | 8 | 1 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 |
| Vu (kN) | 23.4 | 8.27 | 28.91 |
| Tu (kNm) | 0.43 | 0 | 0.43 |
| $V_{Tu}$ (kN) | 2.99 | 0 | 2.99 |
| Vut (kN) | 26.39 | 8.27 | 31.9 |
| Tv (N/sqmm) | 0.31 | 0.1 | 0.37 |
| Tc (N/sqmm) | 0.37 | 0.37 | 0.37 |
| Vc (kN) | 31.88 | 31.88 | 31.88 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0.02 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 93.466 | - | 110.513 |
| Asv Torsion Prv (sqmm) | 365.6 | - | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 140 |
| SvPrv (mm) | 275 | 275 | 140 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 718.14 |

**Maximum Spacing Criteria**

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

**SFR Design**

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group : **G5**

| | | | |
|---|---|---|---|
| Beam No | : | **B15** | |
| Analysis Reference (Member) | : | 160 (11.125 m) | |
| Beam Length | : | 1372 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 | 1 | 1 | 1 |
| Mu (kNm) | - | - | - | 16.45 | 12.92 | 9.82 |
| Tu (kNm) | 10.59 | 10.59 | 10.59 | 10.59 | 10.59 | 10.59 |
| $M_{Tu}$ (kNm) | 17.87 | 17.87 | 17.87 | 17.87 | 17.87 | 17.87 |
| Mud (kNm) | 17.87 | 17.87 | 17.87 | 34.33 | 30.79 | 27.69 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.568 | 0.568 | 0.568 | 1.09 | 0.978 | 0.88 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.241 | 0.215 | 0.2 |
| Pcclc (%) (C) | 0.122 | 0.122 | 0.122 | 0.009 | 0.033 | 0.054 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 204.78 | 182.57 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

| | | | | |
|---|---|---|---|---|
| B' | = | Ptclc | = | Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 |
| Vu (kN) | 11.75 | 7.63 | 2.08 |
| Tu (kNm) | 10.59 | 10.59 | 10.59 |
| $V_{Tu}$ (kN) | 73.66 | 73.66 | 73.66 |
| Vut (kN) | 85.41 | 81.29 | 75.74 |
| Tv (N/sqmm) | 1 | 0.96 | 0.89 |
| Tc (N/sqmm) | 0.37 | 0.37 | 0.37 |
| Vc (kN) | 31.88 | 31.88 | 31.88 |
| Vus=Vut-Vc (kN) | 53.53 | 49.41 | 43.86 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 554.862 | 542.109 | 524.936 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 400.74 | 369.89 | 328.35 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |

Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

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| | | | |
|---|---|---|---|
| Group | : | **G6** | |
| Beam No | : | **B16** | |
| Analysis Reference (Member) | : | 129 (11.125 m) | |
| Beam Length | : | 2134 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 14 | 14 | 14 | 14 | 5 | 5 |
| Critical L/C - RCDC | 10 | 10 | 10 | 10 | 1 | 1 |
| Mu (kNm) | 14.16 | - | - | 2.4 | 52.63 | 82.44 |
| Tu (kNm) | 4.11 | 4.11 | 4.11 | 4.11 | 1.92 | 1.92 |
| $M_{Tu}$ (kNm) | 6.94 | 6.94 | 6.94 | 6.94 | 3.24 | 3.24 |
| Mud (kNm) | 21.1 | 6.94 | 6.94 | 9.35 | 55.87 | 85.67 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.67 | 0.22 | 0.22 | 0.297 | 1.774 | 2.721 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.408 | 0.667 |
| Pcclc (%) (C) | 0 | 0.047 | 0.047 | 0.03 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 346.8 | 567.61 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

| | | | | |
|---|---|---|---|---|
| AstCalc | = | Max {B',C} x B X d (for Mu > 0 ) | | |
| AstCalc | = | Bn x B X d (for Mu = 0) | | |
| Where, | | | | |
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.709 | 0.709 | 0.709 |
| Vu (kN) | 40.69 | 51.23 | 58.24 |
| Tu (kNm) | 1.92 | 1.92 | 1.92 |
| $V_{Tu}$ (kN) | 13.34 | 13.34 | 13.34 |
| Vut (kN) | 54.03 | 64.57 | 71.58 |
| Tv (N/sqmm) | 0.63 | 0.76 | 0.84 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.56 |
| Vc (kN) | 47.85 | 47.85 | 47.85 |
| Vus=Vut-Vc (kN) | 6.18 | 16.72 | 23.73 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 224.747 | 257.741 | 279.666 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |

| | | |
|---|---|---|
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |
| Side Face Reinforcement not required. | | |

---

| | | | |
|---|---|---|---|
| Group | : | **G6** | |
| Beam No | : | **B17** | |
| Analysis Reference (Member) | : | 114 (11.125 m) | |
| Beam Length | : | 4496 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 13 | 13 | 13 | 13 | 20 |
| Critical L/C - RCDC | 16 | 9 | 9 | 9 | 9 | 16 |
| Mu (kNm) | 3.98 | 14.34 | 27.3 | 81.4 | 40.01 | 20.62 |
| Tu (kNm) | 0.13 | 0 | 0 | 0 | 0 | 0.13 |
| $M_{Tu}$ (kNm) | 0.23 | 0 | 0 | 0 | 0 | 0.23 |
| Mud (kNm) | 4.21 | 14.34 | 27.3 | 81.4 | 40.01 | 20.85 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.134 | 0.455 | 0.867 | 2.585 | 1.271 | 0.662 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.627 | 0.283 | 0.2 |
| | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 533.77 | 241.08 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 13 | 13 | 13 |
| Critical L/C - RCDC | 9 | 9 | 9 |
| PtPrv (%) | 0.709 | 0.709 | 0.266 |
| Vu (kN) | 41.05 | 29.8 | 18.56 |
| Tu (kNm) | 0 | 0 | 0 |
| $V_{Tu}$ (kN) | 0 | 0 | 0 |
| Vut (kN) | 41.05 | 29.81 | 18.56 |
| Tv (N/sqmm) | 0.48 | 0.35 | 0.22 |
| Tc (N/sqmm) | 0.56 | 0.56 | 0.37 |
| Vc (kN) | 47.85 | 47.85 | 31.88 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | - | - | - |
| Asv Torsion Prv (sqmm) | - | - | - |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G6** | |
| Beam No | : | **B18** | |
| Analysis Reference (Member) | : | 102 (11.125 m) | |
| Beam Length | : | 5067 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 15 | 13 | 13 | 13 | 18 | 15 |
| Critical L/C - RCDC | 11 | 9 | 9 | 9 | 14 | 11 |
| Mu (kNm) | 9.36 | 14.73 | 14.73 | 36.95 | 8.02 | 27.98 |
| Tu (kNm) | 0.22 | 0.31 | 0.31 | 0.31 | 0.2 | 0.22 |
| $M_{Tu}$ (kNm) | 0.36 | 0.52 | 0.52 | 0.52 | 0.34 | 0.36 |
| Mud (kNm) | 9.72 | 15.25 | 15.25 | 37.47 | 8.36 | 28.34 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.309 | 0.484 | 0.484 | 1.19 | 0.266 | 0.9 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.264 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 224.76 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 5 | 8 | 10 |
| Critical L/C - RCDC | 1 | 4 | 6 |
| PtPrv (%) | 0.709 | 0.266 | 0.709 |
| Vu (kN) | 28.48 | 15.56 | 24.17 |
| Tu (kNm) | 0.32 | 0.29 | 0.22 |
| $V_{Tu}$ (kN) | 2.23 | 2.03 | 1.53 |
| Vut (kN) | 30.71 | 17.6 | 25.7 |
| Tv (N/sqmm) | 0.36 | 0.21 | 0.3 |
| Tc (N/sqmm) | 0.56 | 0.37 | 0.56 |
| Vc (kN) | 47.85 | 31.88 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 105.409 | 63.554 | 86.823 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G6** | |
| Beam No | : | **B19** | |
| Analysis Reference (Member) | : | 90 (11.125 m) | |
| Beam Length | : | 4991 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 13 | 13 | 13 | 13 | 15 |
| Critical L/C - RCDC | 16 | 9 | 9 | 9 | 9 | 11 |
| Mu (kNm) | 8.16 | 20.05 | 20.6 | 43.24 | 10.46 | 26.7 |
| Tu (kNm) | 0.3 | 0.73 | 0.73 | 0.73 | 0.73 | 0.56 |
| $M_{Tu}$ (kNm) | 0.51 | 1.22 | 1.22 | 1.22 | 1.22 | 0.95 |
| Mud (kNm) | 8.67 | 21.28 | 21.82 | 44.47 | 11.69 | 27.65 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.275 | 0.676 | 0.693 | 1.412 | 0.371 | 0.878 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.317 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.709 | 0.709 | 0.709 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 270.11 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 603.18 | 603.18 | 603.18 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T16 | 3-T16 | 3-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 8 | 13 | 15 |
| Critical L/C - RCDC | 4 | 9 | 11 |
| PtPrv (%) | 0.709 | 0.266 | 0.709 |
| Vu (kN) | 31.45 | 15.35 | 23.2 |
| Tu (kNm) | 0.4 | 0.72 | 0.56 |
| $V_{Tu}$ (kN) | 2.82 | 5.05 | 3.91 |
| Vut (kN) | 34.26 | 20.4 | 27.11 |
| Tv (N/sqmm) | 0.4 | 0.24 | 0.32 |
| Tc (N/sqmm) | 0.56 | 0.37 | 0.56 |
| Vc (kN) | 47.85 | 31.88 | 47.85 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 118.98 | 84.892 | 101.173 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G7** | |
| Beam No | : | **B20** | |
| Analysis Reference (Member) | : | 130 (11.125 m) | |
| Beam Length | : | 2134 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 14 | 14 | 14 | 5 | 5 |
| Critical L/C - RCDC | 1 | 10 | 10 | 10 | 1 | 1 |
| Mu (kNm) | 26.7 | - | - | 2.18 | 65.67 | 102.03 |
| Tu (kNm) | 0.17 | 1.99 | 1.99 | 1.99 | 0.17 | 0.17 |
| $M_{Tu}$ (kNm) | 0.29 | 3.36 | 3.36 | 3.36 | 0.29 | 0.29 |
| Mud (kNm) | 27 | 3.36 | 3.36 | 5.54 | 65.96 | 102.32 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |

| | | | | | | |
|---|---|---|---|---|---|---|
| R | 0.857 | 0.107 | 0.107 | 0.176 | 2.095 | 3.25 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.491 | 0.832 |
| Pcclc (%) (C) | 0 | 0.022 | 0.022 | 0.008 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 417.97 | 708.28 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 53.93 | 65.54 | 68.88 |
| Tu (kNm) | 0.18 | 0.18 | 0.18 |
| $V_{Tu}$ (kN) | 1.21 | 1.21 | 1.21 |
| Vut (kN) | 55.15 | 66.76 | 70.1 |
| Tv (N/sqmm) | 0.65 | 0.78 | 0.82 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 10.07 | 13.41 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 179.925 | 216.681 | 227.243 |
| Asv Torsion Prv (sqmm) | 365.6 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 140 | 140 |
| SvPrv (mm) | 275 | 140 | 140 |
| Asv Total Prv (sqmm) | 365.6 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G7** | |
| Beam No | : | **B21** | |
| Analysis Reference (Member) | : | 115 (11.125 m) | |
| Beam Length | : | 4496 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 13 | 13 | 8 | 13 | 20 |
| | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Critical L/C - RCDC | 16 | 9 | 9 | 4 | 9 | 16 |
| Mu (kNm) | 13.74 | 18.71 | 50.43 | 79.46 | 44.71 | 19.69 |
| Tu (kNm) | 0.18 | 0.16 | 0.16 | 0.18 | 0.16 | 0.18 |
| $M_{Tu}$ (kNm) | 0.31 | 0.27 | 0.27 | 0.31 | 0.27 | 0.31 |
| Mud (kNm) | 14.04 | 18.99 | 50.7 | 79.77 | 44.98 | 19.99 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.446 | 0.603 | 1.61 | 2.533 | 1.429 | 0.635 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.366 | 0.612 | 0.321 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.399 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 311.58 | 521.13 | 273.52 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 339.3 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 3-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 8 | 8 | 13 |
| Critical L/C - RCDC | 4 | 4 | 9 |
| PtPrv (%) | 1.107 | 1.107 | 0.399 |
| Vu (kN) | 34.86 | 30.96 | 28.22 |
| Tu (kNm) | 0.18 | 0.18 | 0.16 |
| $V_{Tu}$ (kN) | 1.28 | 1.28 | 1.12 |
| Vut (kN) | 36.15 | 32.25 | 29.34 |
| Tv (N/sqmm) | 0.42 | 0.38 | 0.34 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.45 |
| Vc (kN) | 56.69 | 56.69 | 37.88 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |

| | | | |
|---|---|---|---|
| Asv Torsion (sqmm) | 120.083 | 107.738 | 97.822 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G7** | |
| Beam No | : | **B22** | |
| Analysis Reference (Member) | : | 103 (11.125 m) | |
| Beam Length | : | 5067 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

| **Flexure Design** | | | | | | |
|---|---|---|---|---|---|---|
| | **Beam Bottom** | | | **Beam Top** | | |
| | | | | | | |

| | Left | Mid | Right | Left | Mid | Right |
|---|---|---|---|---|---|---|
| Critical L/C - Analysis | 20 | 13 | 13 | 8 | 13 | 10 |
| Critical L/C - RCDC | 16 | 9 | 9 | 4 | 9 | 6 |
| Mu (kNm) | 19.76 | 14.44 | 29.58 | 38.67 | 15.85 | 20.7 |
| Tu (kNm) | 0.25 | 0.33 | 0.33 | 0.29 | 0.33 | 0.34 |
| $M_{Tu}$ (kNm) | 0.43 | 0.56 | 0.56 | 0.49 | 0.56 | 0.58 |
| Mud (kNm) | 20.19 | 15 | 30.14 | 39.16 | 16.41 | 21.28 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.641 | 0.476 | 0.957 | 1.244 | 0.521 | 0.676 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.21 | 0.277 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 178.55 | 235.62 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 8 | 8 | 10 |
| Critical L/C - RCDC | 4 | 4 | 6 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 21.58 | 16.27 | 15 |
| Tu (kNm) | 0.29 | 0.29 | 0.34 |
| $V_{Tu}$ (kN) | 2.01 | 2.01 | 2.38 |
| Vut (kN) | 23.58 | 18.27 | 17.37 |
| Tv (N/sqmm) | 0.28 | 0.21 | 0.2 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |

| | | | |
|---|---|---|---|
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 83.503 | 66.702 | 65.501 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G7** | |
| Beam No | : | **B23** | |
| Analysis Reference (Member) | : | 91 (11.125 m) | |
| Beam Length | : | 4991 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 8 | 13 | 8 | 13 | 20 |
| Critical L/C - RCDC | 16 | 4 | 9 | 4 | 9 | 16 |
| Mu (kNm) | 16.97 | 19.35 | 35.45 | 39.05 | 14.66 | 20.02 |
| Tu (kNm) | 0 | 0 | 0 | 0 | 0 | 0 |
| $M_{Tu}$ (kNm) | 0 | 0 | 0 | 0 | 0 | 0 |
| Mud (kNm) | 16.97 | 19.35 | 35.45 | 39.05 | 14.66 | 20.02 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.539 | 0.614 | 1.126 | 1.24 | 0.466 | 0.636 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.249 | 0.276 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 211.9 | 234.91 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | **Left** | **Mid** | **Right** |
|---|---|---|---|
| Critical L/C - Analysis | 8 | 8 | 10 |
| Critical L/C - RCDC | 4 | 4 | 6 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 23.05 | 17.74 | 13.8 |
| Tu (kNm) | 0 | 0 | 0 |
| $V_{Tu}$ (kN) | 0 | 0 | 0 |
| Vut (kN) | 23.05 | 17.74 | 13.8 |
| Tv (N/sqmm) | 0.27 | 0.21 | 0.16 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |

| | | | |
|---|---|---|---|
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | - | - | - |
| Asv Torsion Prv (sqmm) | - | - | - |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G8** | |
| Beam No | : | **B24** | |
| Analysis Reference (Member) | : | 131 (11.125 m) | |
| Beam Length | : | 2134 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Cantilever Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

| Flexure Design | | | | | | |
|---|---|---|---|---|---|---|
| | Beam Bottom | | | Beam Top | | |
| | Left | Mid | Right | Left | Mid | Right |
| Critical L/C - Analysis | 12 | 12 | 12 | 5 | 5 | 5 |
| Critical L/C - RCDC | 8 | 8 | 8 | 1 | 1 | 1 |
| Mu (kNm) | 9.89 | - | - | 11.91 | 63.81 | 93.65 |
| Tu (kNm) | 7.33 | 7.33 | 7.33 | 6.6 | 6.6 | 6.6 |
| $M_{Tu}$ (kNm) | 12.37 | 12.37 | 12.37 | 11.14 | 11.14 | 11.14 |
| Mud (kNm) | 22.26 | 12.37 | 12.37 | 23.04 | 74.95 | 104.79 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.707 | 0.393 | 0.393 | 0.732 | 2.38 | 3.328 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.569 | 0.852 |
| Pcclc (%) (C) | 0.016 | 0.084 | 0.084 | 0 | 0 | 0.022 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 484.24 | 725.25 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| Shear Design | | | |
|---|---|---|---|
| | Left | Mid | Right |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 44.22 | 53.09 | 56.66 |
| Tu (kNm) | 6.6 | 6.6 | 6.6 |
| $V_{Tu}$ (kN) | 45.89 | 45.89 | 45.89 |
| Vut (kN) | 90.12 | 98.98 | 102.55 |
| | | | |

| | | | |
|---|---|---|---|
| Tv (N/sqmm) | 1.06 | 1.16 | 1.21 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 33.43 | 42.29 | 45.86 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 488.023 | 516.089 | 527.371 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 316.64 | 343.32 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

**SFR Design**

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G8** | |
| Beam No | : | **B25** | |
| Analysis Reference (Member) | : | 116 (11.125 m) | |
| Beam Length | : | 4496 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |

| | | | |
|---|---|---|---|
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 13 | 13 | 8 | 13 | 20 |
| Critical L/C - RCDC | 16 | 9 | 9 | 4 | 9 | 16 |
| Mu (kNm) | 11.59 | 14.15 | 39.24 | 61.9 | 36.02 | 17.79 |
| Tu (kNm) | 0.44 | 0.52 | 0.52 | 0.38 | 0.52 | 0.44 |
| $M_{Tu}$ (kNm) | 0.75 | 0.88 | 0.88 | 0.64 | 0.88 | 0.75 |
| Mud (kNm) | 12.33 | 15.04 | 40.12 | 62.54 | 36.91 | 18.54 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.392 | 0.478 | 1.274 | 1.986 | 1.172 | 0.589 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.284 | 0.462 | 0.26 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.399 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 241.83 | 393.47 | 221.16 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 339.3 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 3-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | **Left** | **Mid** | **Right** |
|---|---|---|---|
| Critical L/C - Analysis | 8 | 13 | 13 |

| | | | |
|---|---|---|---|
| Critical L/C - RCDC | 4 | 9 | 9 |
| PtPrv (%) | 1.107 | 1.107 | 0.399 |
| Vu (kN) | 26.86 | 22.33 | 22.33 |
| Tu (kNm) | 0.38 | 0.52 | 0.52 |
| $V_{Tu}$ (kN) | 2.65 | 3.63 | 3.63 |
| Vut (kN) | 29.51 | 25.96 | 25.96 |
| Tv (N/sqmm) | 0.35 | 0.31 | 0.31 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.45 |
| Vc (kN) | 56.69 | 56.69 | 37.88 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 105.115 | 98.231 | 98.231 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

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| | | | |
|---|---|---|---|
| Group | : | **G8** | |
| Beam No | : | **B26** | |
| Analysis Reference (Member) | : | 104 (11.125 m) | |
| Beam Length | : | 5067 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |

| | | | |
|---|---|---|---|
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 20 | 13 | 13 | 8 | 13 | 10 |
| Critical L/C - RCDC | 16 | 9 | 9 | 4 | 9 | 6 |
| Mu (kNm) | 22.33 | 16.05 | 31.26 | 33.88 | 14.36 | 22.9 |
| Tu (kNm) | 1.42 | 2.64 | 2.64 | 2.24 | 2.64 | 2.07 |
| $M_{Tu}$ (kNm) | 2.39 | 4.45 | 4.45 | 3.78 | 4.45 | 3.5 |
| Mud (kNm) | 24.72 | 20.5 | 35.7 | 37.65 | 18.81 | 26.39 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.785 | 0.651 | 1.134 | 1.196 | 0.597 | 0.838 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.251 | 0.265 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 213.52 | 225.94 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

**Shear Design**

| | Left | Mid | Right |
|---|---|---|---|
| Critical L/C - Analysis | 8 | 8 | 10 |
| Critical L/C - RCDC | 4 | 4 | 6 |
| PtPrv (%) | 1.107 | 1.107 | 1.107 |
| Vu (kN) | 18.75 | 14.43 | 13.93 |
| Tu (kNm) | 2.24 | 2.24 | 2.07 |
| $V_{Tu}$ (kN) | 15.56 | 15.56 | 14.4 |
| Vut (kN) | 34.31 | 30 | 28.34 |
| Tv (N/sqmm) | 0.4 | 0.35 | 0.33 |
| Tc (N/sqmm) | 0.67 | 0.67 | 0.67 |
| Vc (kN) | 56.69 | 56.69 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 177.37 | 163.706 | 153.337 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G8** | |
| Beam No | : | **B27** | |
| Analysis Reference (Member) | : | 92 (11.125 m) | |
| Beam Length | : | 4991 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |

| | | | |
|---|---|---|---|
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 15 | 13 | 13 | 13 | 18 | 15 |
| Critical L/C - RCDC | 11 | 9 | 9 | 9 | 14 | 11 |
| Mu (kNm) | 15.47 | 23.72 | 23.72 | 50.96 | 11.56 | 42.46 |
| Tu (kNm) | 0.66 | 1.02 | 1.02 | 1.02 | 0.68 | 0.66 |
| $M_{Tu}$ (kNm) | 1.11 | 1.72 | 1.72 | 1.72 | 1.15 | 1.11 |
| Mud (kNm) | 16.59 | 25.44 | 25.44 | 52.69 | 12.72 | 43.57 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.527 | 0.808 | 0.808 | 1.673 | 0.404 | 1.384 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.382 | 0.2 | 0.311 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 1.107 | 1.107 | 1.107 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 325.03 | 170.2 | 264.24 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 942.48 | 942.48 | 942.48 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 3-T20 | 3-T20 | 3-T20 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 8 | 8 | 10 |
| Critical L/C - RCDC | 4 | 4 | 6 |
| PtPrv (%) | 1.107 | 0.266 | 1.107 |
| Vu (kN) | 37.76 | 22.6 | 32.13 |
| Tu (kNm) | 0.84 | 0.84 | 0.55 |
| $V_{Tu}$ (kN) | 5.84 | 5.84 | 3.83 |
| Vut (kN) | 43.6 | 28.44 | 35.96 |
| Tv (N/sqmm) | 0.51 | 0.33 | 0.42 |
| Tc (N/sqmm) | 0.67 | 0.37 | 0.67 |
| Vc (kN) | 56.69 | 31.88 | 56.69 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 163.783 | 115.79 | 130.768 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

---

| | | | |
|---|---|---|---|
| Group | : | **G9** | |
| Beam No | : | **B28** | |
| Analysis Reference (Member) | : | 161 (11.125 m) | |
| Beam Length | : | 2134 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |

| | | | |
|---|---|---|---|
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 | 1 | 1 | 1 |
| Mu (kNm) | - | - | - | 12.95 | 27.68 | 40.05 |
| Tu (kNm) | 9.82 | 9.82 | 9.82 | 9.82 | 9.82 | 9.82 |
| $M_{Tu}$ (kNm) | 16.58 | 16.58 | 16.58 | 16.58 | 16.58 | 16.58 |
| Mud (kNm) | 16.58 | 16.58 | 16.58 | 29.53 | 44.26 | 56.63 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.526 | 0.526 | 0.526 | 0.938 | 1.406 | 1.798 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.205 | 0.316 | 0.414 |
| Pcclc (%) (C) | 0.113 | 0.113 | 0.113 | 0.024 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.473 | 0.473 | 0.473 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 174.72 | 268.76 | 352.04 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 402.12 | 402.12 | 402.12 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T16 | 2-T16 | 2-T16 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

| | | | | |
|---|---|---|---|---|
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 5 |
| Critical L/C - RCDC | 1 | 1 | 1 |
| PtPrv (%) | 0.473 | 0.473 | 0.473 |
| Vu (kN) | 9.31 | 18.31 | 25.54 |
| Tu (kNm) | 9.82 | 9.82 | 9.82 |
| $V_{Tu}$ (kN) | 68.31 | 68.31 | 68.31 |
| Vut (kN) | 77.62 | 86.62 | 93.85 |
| Tv (N/sqmm) | 0.91 | 1.02 | 1.1 |
| Tc (N/sqmm) | 0.48 | 0.48 | 0.48 |
| Vc (kN) | 40.65 | 40.65 | 40.65 |
| Vus=Vut-Vc (kN) | 36.97 | 45.97 | 53.2 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 528.03 | 556.192 | 578.828 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 276.76 | 344.11 | 398.26 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

Group : **G9**

| | | | |
|---|---|---|---|
| Beam No | : | **B29** | |
| Analysis Reference (Member) | : | 162 (11.125 m) | |
| Beam Length | : | 4496 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 12 | 5 | 13 | 12 |
| Critical L/C - RCDC | 8 | 8 | 8 | 1 | 9 | 8 |
| Mu (kNm) | - | - | - | 41.85 | 20.08 | 15.7 |
| Tu (kNm) | 1.54 | 1.54 | 1.54 | 0.8 | 0.72 | 1.54 |
| $M_{Tu}$ (kNm) | 2.6 | 2.6 | 2.6 | 1.35 | 1.22 | 2.6 |
| Mud (kNm) | 2.6 | 2.6 | 2.6 | 43.2 | 21.3 | 18.29 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.082 | 0.082 | 0.082 | 1.372 | 0.676 | 0.581 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.308 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0.017 | 0.017 | 0.017 | 0 | 0 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.473 | 0.473 | 0.266 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 261.84 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 402.12 | 402.12 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T16 | 2-T16 | 2-T12 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

| | | | | |
|---|---|---|---|---|
| B' | = | Ptclc | = | Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 5 | 15 |
| Critical L/C - RCDC | 1 | 1 | 11 |
| PtPrv (%) | 0.473 | 0.473 | 0.266 |
| Vu (kN) | 25.43 | 12.4 | 14.95 |
| Tu (kNm) | 0.8 | 0.8 | 1.14 |
| $V_{Tu}$ (kN) | 5.58 | 5.58 | 7.92 |
| Vut (kN) | 31.02 | 17.99 | 22.87 |
| Tv (N/sqmm) | 0.36 | 0.21 | 0.27 |
| Tc (N/sqmm) | 0.48 | 0.48 | 0.37 |
| Vc (kN) | 40.65 | 40.65 | 31.88 |
| Vus=Vut-Vc (kN) | 0 | 0 | 0 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 120.366 | 79.592 | 102.025 |
| Asv Torsion Prv (sqmm) | 365.6 | 365.6 | 365.6 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 275 | 275 | 275 |
| SvPrv (mm) | 275 | 275 | 275 |
| Asv Total Prv (sqmm) | 365.6 | 365.6 | 365.6 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

| | | | |
|---|---|---|---|
| Group | : | **G9** | |
| Beam No | : | **B30** | |
| Analysis Reference (Member) | : | 163 (11.125 m) | |
| Beam Length | : | 5067 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 5 | 13 | 12 | 12 | 15 |
| Critical L/C - RCDC | 8 | 1 | 9 | 8 | 8 | 11 |
| Mu (kNm) | - | 10.54 | 6.68 | 20.29 | 0.04 | 19.95 |
| Tu (kNm) | 5.04 | 3.8 | 3.72 | 5.04 | 5.04 | 3.47 |
| $M_{Tu}$ (kNm) | 8.5 | 6.41 | 6.29 | 8.5 | 8.5 | 5.85 |
| Mud (kNm) | 8.5 | 16.95 | 12.96 | 28.8 | 8.54 | 25.8 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.27 | 0.538 | 0.412 | 0.915 | 0.271 | 0.819 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Pcclc (%) (C) | 0.057 | 0 | 0 | 0 | 0.057 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 |
| AstCalc (sqmm) | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 | 170.2 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 |

**Note: Calculation of Ast**

AstCalc = Max {B',C} x B X d (for Mu > 0 )
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

| | | | | |
|---|---|---|---|---|
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 12 | 12 | 5 |
| Critical L/C - RCDC | 8 | 8 | 1 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 |
| Vu (kN) | 8.07 | 8.07 | 21.19 |
| Tu (kNm) | 5.04 | 5.04 | 3.8 |
| $V_{Tu}$ (kN) | 35.04 | 35.04 | 26.43 |
| Vut (kN) | 43.12 | 43.12 | 47.62 |
| Tv (N/sqmm) | 0.51 | 0.51 | 0.56 |
| Tc (N/sqmm) | 0.37 | 0.37 | 0.37 |
| Vc (kN) | 31.88 | 31.88 | 31.88 |
| Vus=Vut-Vc (kN) | 11.24 | 11.24 | 15.74 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 271.644 | 271.644 | 251.59 |
| Asv Torsion Prv (sqmm) | 718.14 | 718.14 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 140 | 140 |
| SvPrv (mm) | 140 | 140 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 718.14 | 718.14 |

**Maximum Spacing Criteria**

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

**SFR Design**

Beam Width = 230 mm

| | | |
|---|---|---|
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |
| Side Face Reinforcement not required. | | |

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| | | | |
|---|---|---|---|
| Group | : | **G9** | |
| Beam No | : | **B31** | |
| Analysis Reference (Member) | : | 164 (11.125 m) | |
| Beam Length | : | 4991 | mm |
| Breadth (B) | : | 230 | mm |
| Depth (D) | : | 430 | mm |
| Effective Depth (d) | : | 370 | mm |
| Design Code | : | IS 456 | |
| Beam Type | : | Regular Beam | |
| Grade Of Concrete (Fck) | : | M25 | N/sqmm |
| Grade Of Steel (Main) | : | Fe550 | N/sqmm |
| Grade Of Steel (Shear) | : | Fe550 | N/sqmm |
| Top/Bottom Clear Cover (Cmin) | : | 30 | mm |
| Side Clear Cover | : | 30 | mm |
| Es | : | 2x10^5 | N/sqmm |
| Ptnominal (%) (Bn) | : | 0.2 | % |

**Flexure Design**

| | Beam Bottom | | | Beam Top | | |
|---|---|---|---|---|---|---|
| | **Left** | **Mid** | **Right** | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 15 | 5 | 13 | 13 | 13 | 15 |
| Critical L/C - RCDC | 11 | 1 | 9 | 9 | 9 | 11 |
| Mu (kNm) | 5.15 | 31.92 | 13.74 | 34.82 | - | 29.47 |
| Tu (kNm) | 0.73 | 1.25 | 1.42 | 1.42 | 1.42 | 0.73 |
| $M_{Tu}$ (kNm) | 1.23 | 2.1 | 2.39 | 2.39 | 2.39 | 1.23 |
| Mud (kNm) | 6.38 | 34.03 | 16.13 | 37.22 | 2.39 | 30.7 |
| MuLim (kNm) | 102 | 102 | 102 | 102 | 102 | 102 |
| R | 0.203 | 1.081 | 0.512 | 1.182 | 0.076 | 0.975 |
| Ptmin (%) (A) | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Ptclc (%) (B') | 0.2 | 0.238 | 0.2 | 0.262 | 0.2 | 0.214 |
| Pcclc (%) (C) | 0 | 0 | 0 | 0 | 0.016 | 0 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 | 0.266 |
| AstCalc (sqmm) | 170.2 | 202.9 | 170.2 | 223.14 | 170.2 | 182.01 |
| AstPrv (sqmm) | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 | 226.2 |
| Reinforcement Provided | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 | 2-T12 |

**Note: Calculation of Ast**

| | | | | |
|---|---|---|---|---|
| AstCalc | = | Max {B',C} x B X d (for Mu > 0 ) | | |
| AstCalc | = | Bn x B X d (for Mu = 0) | | |
| Where, | | | | |
| A | = | Ptmin | = | Min % area of flexural reinforcement |
| B' | = | Ptclc | = | Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A) |
| Bn | = | Ptnominal | = | Nominal % area of reinforcement |
| C | = | Pcclc | = | % area of compression reinforcement required |
| D | = | AstCrack | = | Area of reinforcement required for Crackwidth |

| **Shear Design** | | | |
|---|---|---|---|
| | **Left** | **Mid** | **Right** |
| Critical L/C - Analysis | 5 | 13 | 5 |
| Critical L/C - RCDC | 1 | 9 | 1 |
| PtPrv (%) | 0.266 | 0.266 | 0.266 |
| Vu (kN) | 46.26 | 19.28 | 42.15 |
| Tu (kNm) | 1.25 | 1.42 | 1.25 |
| $V_{Tu}$ (kN) | 8.68 | 9.86 | 8.68 |
| Vut (kN) | 54.94 | 29.13 | 50.82 |
| Tv (N/sqmm) | 0.65 | 0.34 | 0.6 |
| Tc (N/sqmm) | 0.37 | 0.37 | 0.37 |
| Vc (kN) | 31.88 | 31.88 | 31.88 |
| Vus=Vut-Vc (kN) | 23.06 | 0 | 18.94 |
| Legs | 2 | 2 | 2 |
| Stirrup Rebar | 8 | 8 | 8 |
| Asv Torsion (sqmm) | 204.236 | 129.025 | 191.496 |
| Asv Torsion Prv (sqmm) | 718.14 | 365.6 | 718.14 |
| Asv Reqd (sqmm/m) | 254.81 | 254.81 | 254.81 |
| SvCalc (mm) | 140 | 275 | 140 |
| SvPrv (mm) | 140 | 275 | 140 |
| Asv Total Prv (sqmm) | 718.14 | 365.6 | 718.14 |

**Maximum Spacing Criteria**

Basic

| | | |
|---|---|---|
| Spc1 =0.75d | = 278 | mm |
| Spc2 | = 300 | mm |

For Torsion

(X1 = 180, Y1 = 380)

| | | |
|---|---|---|
| Spc3 = X1 | = 180 | mm |

| | | |
|---|---|---|
| Spc4=(X1+Y1)/4 | = 140 | mm |

**SFR Design**

| | | |
|---|---|---|
| Beam Width | = 230 | mm |
| Beam Depth | = 430 | mm |
| Web Depth | = 430 <= 750 | mm |

Side Face Reinforcement not required.

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